

CORE PURE / MISCELLANEOUS PURE

# Roots of Polynomials

Questions by topic

15 questions · 117 marks

## Questions

|             |          |    |
|-------------|----------|----|
| Question 1  | 10 marks | 2  |
| Question 2  | 8 marks  | 3  |
| Question 3  | 4 marks  | 4  |
| Question 4  | 12 marks | 5  |
| Question 5  | 7 marks  | 6  |
| Question 6  | 7 marks  | 7  |
| Question 7  | 9 marks  | 8  |
| Question 8  | 8 marks  | 9  |
| Question 9  | 5 marks  | 10 |
| Question 10 | 8 marks  | 11 |
| Question 11 | 6 marks  | 12 |
| Question 12 | 8 marks  | 13 |
| Question 13 | 10 marks | 14 |
| Question 14 | 6 marks  | 15 |
| Question 15 | 9 marks  | 16 |

1. The cubic equation

$$x^3 - 5x^2 + px - q = 0$$

where  $p$  and  $q$  are positive real constants, has roots  $\alpha$ ,  $\beta$  and  $\gamma$ .

Given that

$$(\alpha - \beta)^2 + (\beta - \gamma)^2 + (\gamma - \alpha)^2 = 14$$

(a) show that  $p = 6$

[3]

Given that

$$\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} = 3$$

(b) determine the value of  $q$

[3]

(c) Without solving the cubic equation, determine the value of  $(\alpha + 1)(\beta + 1)(\gamma + 1)$

[4]

**2.** The quartic equation  $x^4 + bx^3 + cx^2 + dx + 3 = 0$  has roots  $\alpha, \beta, \gamma, \delta$ . It is given that

$$\alpha + \beta + \gamma + \delta = -2, \quad (\alpha + 1)^2 + (\beta + 1)^2 + (\gamma + 1)^2 + (\delta + 1)^2 = 6, \quad \alpha^{-1} + \beta^{-1} + \gamma^{-1} + \delta^{-1} = 2$$

(a) Find the values of  $b, c$  and  $d$  [6]

(b) Given also that

$$(\alpha + 1)^3 + (\beta + 1)^3 + (\gamma + 1)^3 + (\delta + 1)^3 = 20$$

find the value of  $\alpha^4 + \beta^4 + \gamma^4 + \delta^4$  [2]

- 3.** The equation  $x^3 - 4x^2 - x + 6 = 0$  has roots  $\alpha$ ,  $\beta$  and  $\gamma$ .

Determine the value of

$$\frac{1}{\alpha + \beta} + \frac{1}{\beta + \gamma} + \frac{1}{\gamma + \alpha}$$

[4]

4. The quartic equation  $3x^4 - x^3 - 10x^2 + 8x + 4 = 0$  has roots  $\alpha$ ,  $\beta$ ,  $\gamma$  and  $\delta$ .

Without solving the equation, find equations with integer coefficients whose roots are

(a)  $\frac{1}{\alpha}$ ,  $\frac{1}{\beta}$ ,  $\frac{1}{\gamma}$  and  $\frac{1}{\delta}$  [6]

(b)  $\frac{2}{\alpha-1}$ ,  $\frac{2}{\beta-1}$ ,  $\frac{2}{\gamma-1}$  and  $\frac{2}{\delta-1}$  [6]

5. The cubic equation  $x^3 + x^2 - 2x - 1 = 0$  has roots  $\alpha$ ,  $\beta$  and  $\gamma$ .

(a) Find a cubic equation whose roots are  $\frac{1}{\alpha+1}$ ,  $\frac{1}{\beta+1}$  and  $\frac{1}{\gamma+1}$  [3]

(b) Find the value of

$$\left(\frac{1}{\alpha+1}\right)^2 + \left(\frac{1}{\beta+1}\right)^2 + \left(\frac{1}{\gamma+1}\right)^2$$
 [2]

(c) Find the value of

$$\left(\frac{1}{\alpha+1}\right)^3 + \left(\frac{1}{\beta+1}\right)^3 + \left(\frac{1}{\gamma+1}\right)^3$$
 [2]

**6.** The equation  $x^3 - 14x^2 + 56x + c = 0$ , where  $c$  is a constant, has three roots in geometric progression.

(a) Determine the roots of the equation [6]

(b) Find  $c$  [1]

**7.** The quadratic equation

$$x^2 - kx + 1 = 0$$

where  $k$  is an integer, has roots  $\alpha$  and  $\beta$ .

(a) Write down, in terms of  $k$  where appropriate, the value of  $\alpha + \beta$  and the value of  $\alpha\beta$  [2]

(b) Determine, in simplest form in terms of  $k$ , the value of

$$(\alpha^2 + \beta) + (\beta^2 + \alpha) \quad [3]$$

(c) Determine a quadratic equation which has roots

$$\alpha^2 + \beta \text{ and } \beta^2 + \alpha$$

giving your answer in the form  $px^2 + qx + r = 0$ , where  $p$ ,  $q$  and  $r$  are integer values in terms of  $k$  [4]

8. The roots of the equation

$$x^3 + 5x^2 - 4x - 24 = 0$$

are  $\alpha$ ,  $\beta$  and  $\gamma$ .

Without solving the equation,

(a) write down the value of each of

$$\alpha + \beta + \gamma \quad \alpha\beta + \beta\gamma + \gamma\alpha \quad \alpha\beta\gamma \quad [1]$$

(b) Hence determine the value of

(i)

$$\frac{1}{\alpha\beta} + \frac{1}{\beta\gamma} + \frac{1}{\gamma\alpha} \quad [2]$$

(ii)

$$(\alpha + 2)(\beta + 2)(\gamma + 2) \quad [2]$$

(iii)

$$\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2 \quad [3]$$

**9.** The roots of the equation  $3x^3 + px^2 - 12x - 8 = 0$  are  $\alpha$ ,  $\beta$  and  $\gamma$ .

(a) Given that  $\alpha + \beta + \gamma = 2$ , write down the value of  $p$  [1]

(b) Write down values for  $\alpha\beta + \beta\gamma + \gamma\alpha$  and  $\alpha\beta\gamma$  [1]

(c) Hence find the value of  $\left(1 + \frac{1}{\alpha}\right) \left(1 + \frac{1}{\beta}\right) \left(1 + \frac{1}{\gamma}\right)$  [3]

**10.** The roots of the equation  $x^3 + 2x^2 - 5x - 3 = 0$  are  $\alpha$ ,  $\beta$  and  $\gamma$ .

(a) Find  $\alpha^2\beta^2 + \beta^2\gamma^2 + \gamma^2\alpha^2$  [4]

(b) Find an equation with integer coefficients whose roots are  $\alpha^2$ ,  $\beta^2$  and  $\gamma^2$  [4]

- 11.** The graph of  $y = x^3 + 3x^2 + px - 20$ , where  $p$  is a constant, crosses the  $x$ -axis at three points whose  $x$ -coordinates are equally spaced.

Find the roots of

$$x^3 + 3x^2 + px - 20 = 0$$

[6]

**12.** The cubic equation  $x^3 - 3x + 1 = 0$  has roots  $\alpha$ ,  $\beta$  and  $\gamma$ .

(a) Find a cubic equation whose roots are  $\frac{1}{\alpha-1}$ ,  $\frac{1}{\beta-1}$ ,  $\frac{1}{\gamma-1}$ . [3]

(b) Hence find the value of  $\frac{1}{(\alpha-1)^2} + \frac{1}{(\beta-1)^2} + \frac{1}{(\gamma-1)^2}$ . [2]

(c) Find also the value of  $\frac{1}{(\alpha-1)^5} + \frac{1}{(\beta-1)^5} + \frac{1}{(\gamma-1)^5}$ . [3]

**13.** The quartic equation  $x^4 - 2x^2 + 2x + 1 = 0$  has roots  $\alpha, \beta, \gamma, \delta$ .

(a) Find a quartic equation whose roots are  $\alpha^2, \beta^2, \gamma^2, \delta^2$  and state the value of  $\alpha^2 + \beta^2 + \gamma^2 + \delta^2$ . [5]

(b) Find the value of  $\alpha^6 + \beta^6 + \gamma^6 + \delta^6$ . [3]

(c) Find the value of  $\alpha^8 + \beta^8 + \gamma^8 + \delta^8$ . [2]

**14.** The equation  $z^4 - 2z^3 - z^2 - 2z + 1 = 0$  has roots  $\alpha$ ,  $\beta$ ,  $\gamma$  and  $\delta$ .

(a) Show that a quartic equation whose roots are  $\alpha + \frac{1}{\alpha}$ ,  $\beta + \frac{1}{\beta}$ ,  $\gamma + \frac{1}{\gamma}$  and  $\delta + \frac{1}{\delta}$  is

$$w^4 - 4w^3 - 2w^2 + 12w + 9 = 0 \quad [3]$$

(b) Hence determine the exact roots of the equation  $z^4 - 2z^3 - z^2 - 2z + 1 = 0$ . [3]

**15.** The quartic equation  $x^4 - 2x^3 + 3x^2 - 4x + 1 = 0$  has roots  $\alpha, \beta, \gamma, \delta$ .

(a) Find the value of  $\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}$ . [2]

(b) Find the value of  $\frac{1}{\alpha^2} + \frac{1}{\beta^2} + \frac{1}{\gamma^2} + \frac{1}{\delta^2}$ . [2]

(c) Find the value of

$$\left(\frac{1}{\alpha} + \frac{1}{\beta} + \frac{1}{\gamma}\right)^2 + \left(\frac{1}{\beta} + \frac{1}{\gamma} + \frac{1}{\delta}\right)^2 + \left(\frac{1}{\gamma} + \frac{1}{\delta} + \frac{1}{\alpha}\right)^2 + \left(\frac{1}{\delta} + \frac{1}{\alpha} + \frac{1}{\beta}\right)^2$$
 [5]